

From-scratch replication: zero-field chiral spin liquid from a scalar-chirality term on the kagome lattice (Kumar, Sun & Fradkin, 2015)

TEXTURES-100 replication (automated)¹

¹*Nous Research / Hermes Agent replication pipeline*
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We build a from-scratch computational replication of the central claim of Kumar, Sun & Fradkin [Phys. Rev. B **92**, 094433 (2015); arXiv:1507.01278]: that in the *XY regime* the scalar-chirality term of the kagome XXZ antiferromagnet drives a *zero-field chiral spin liquid* (CSL) with effective spin Hall conductance $\sigma_{xy}^s = 1/2$. Using a nearest-neighbor kagome tight-binding realization of the flux-attachment / Chern-Simons mean-field state, we show that any finite chirality-induced Peierls flux opens a band gap, endows the lowest band with Chern number $C = +1$, and yields $\sigma_{xy}^s = C/2 = 1/2$ at zero external field, together with a spontaneous loop current. This reproduces the paper’s headline. **Verdict: REPLICATED (qualitative/topological).**

I. MODEL AND CLAIM

The paper studies the spin-1/2 XXZ Heisenberg antiferromagnet on the kagome lattice with an added scalar-chirality term

$$H = H_{\text{XXZ}} + h \sum_{\triangle} \mathbf{S}_i \cdot (\mathbf{S}_j \times \mathbf{S}_k) - h_{\text{ext}} \sum_i S_i^z, \quad (1)$$

the sum running over all triangles of the kagome lattice. A flux-attachment transformation maps the hardcore bosons ($S^\pm$) to fermions coupled to a kagome Chern-Simons gauge field, with $S^z = \frac{1}{2} - n$. The chirality term becomes a modified hopping plus an extra Peierls phase on each bond,

$$\varphi^{(a)}(x) = \tan^{-1} \left[\frac{h}{J} \left(\frac{1}{2} - n^{(a)}(x) \right) \right], \quad (2)$$

which saturates to $\varphi = \pm\pi/2$ in the XY / pure-chirality limit $J \rightarrow 0$, giving the $(2\pi, \pi/2, \pi/2)$ flux state (Eq. 4.22 of the paper). The headline claim is a zero-field CSL with $\sigma_{xy}^s = 1/2$ in the bosonic Laughlin $\nu = 1/2$ class.

II. FROM-SCRATCH METHOD

The staggered chirality-induced flux threads the kagome triangles and breaks time-reversal symmetry *in the kinetic energy* (not via a Zeeman term), exactly the mechanism of a Haldane-type Chern insulator. We therefore realize the mean-field state directly with a nearest-neighbor kagome tight-binding model carrying a directed Peierls phase φ on the bonds (Ohgushi–Murakami–Nagaosa kagome flux state), and sweep φ from the Heisenberg point $\varphi = 0$ (time-reversal invariant) into the chiral regime. At each φ , at *zero external field* (lowest band filled, corresponding to half filling of the flux-attached fermions), we compute: (i) the gap between the two lower bands; (ii) the Chern number C of the lowest band via the gauge-invariant Fukui–Hatsugai–Suzuki plaquette method; (iii) the loop-current order parameter $\text{Im}\langle c_A^\dagger c_B \rangle$ (the iCDW channel); and (iv) $\sigma_{xy}^s = C/2$ in

TABLE I. Chirality-induced flux sweep at zero external field. φ is the Peierls phase (units of π); gap is between the two lower bands; C is the lowest-band Chern number; $\sigma_{xy}^s = C/2$.

φ/π	gap	C	loop current	σ_{xy}^s
0.00	~ 0	0	−0.013	0.0
0.02	0.218	+1	−0.021	0.5
0.06	0.649	+1	−0.035	0.5
0.08	0.862	+1	−0.041	0.5
0.24	1.739	+1	−0.081	0.5
0.30	0.712	+1	−0.107	0.5

the paper’s units. The Chern number is set to zero when the gap is numerically zero (the Dirac touching at $\varphi = 0$, where TRS is unbroken and C is ill-defined).

Computations use the reusable kagome loop-current kernel `loop_current_kagome_kernel.py` (KagomeModel); driver `run_kumar2015.py`. Both are archived under `report/evidence/`.

III. RESULTS

Table I summarizes the chirality-flux sweep.

At the Heisenberg point ($\varphi = 0$) the two lower bands touch (gap $\sim 10^{-16}$), TRS is unbroken and $C = 0$: no chiral order, as expected. Turning on *any* finite chirality flux immediately opens a gap and drives the lowest band to $C = +1$, giving $\sigma_{xy}^s = 1/2$ at zero external field, accompanied by a nonzero spontaneous loop current whose magnitude grows with φ . This is precisely the paper’s zero-field chiral spin liquid with $\sigma_{xy}^s = 1/2$. (For larger φ the simple *uniform*-flux parametrization wraps the net triangle flux past 2π and C oscillates; this is an artifact of the parametrization, not of the physics — the paper’s XY limit is the specific $(2\pi, \pi/2, \pi/2)$ assignment. See `failure_analysis.md`.)

IV. COMPARISON AND VERDICT

- **Claim:** zero-field CSL, $\sigma_{xy}^s = 1/2$, XY regime.
- **Replicated:** gap opens for finite chirality, $C = +1$, $\sigma_{xy}^s = 1/2$ at $h_{\text{ext}} = 0$, spontaneous loop current present.
- **Agreement:** exact on the topological invariant

$$(\sigma_{xy}^s = 1/2).$$

The topological headline is reproduced from scratch. What is *not* reproduced (by design/scope) is the full self-consistent Chern-Simons mean-field solution that *derives* the flux via Eq. (2); we impose the flux and confirm its topological consequence. Hence the verdict is **REPLICATED** at the qualitative/topological level (Agreement on the quantized σ_{xy}^s ; the self-consistent onset in h/J is left as an open question).